AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

FIN302-F
Portfolio Theory

Final Examination / Version: B
2025-2026 / Spring

13 – June – 2026

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed: 2 hours**

Instructions:

1- Take **FIVE (5)** minutes to read all instructions carefully before attempting to write.
2- Mobile phones must be in "SWITCHED **OFF"** mode throughout the examination.
3- This is a **CLOSED** book examination. NO documents of any kind are allowed.
4- You **MUST** answer the questions using only the answer booklets provided
5- You **MUST** return the examination paper together with your answer booklets.
6- Do **NOT** write your answers in pencil except graphs/sketches
7- Students are **NOT** allowed to use any type of dictionary or mobile phones during the examination
8- Students are allowed to use ***non-programmable*** calculators only (if needed).
9- Any form of cheating will **NOT** be tolerated and will subject you to AOU Cheating Policy.

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Question 1: Choose the right answer for each of the following questions (25 marks; 5 marks per question)

1. Martin's Inc. is expected to pay annual dividends of $2.50 a share for the next three years. After that, dividends are expected to increase by 3% annually. What is the current value of this stock to you if you require a 9% rate of return on this investment?

A) $39.47
B) $40.11
C) $41.81
D) $42.92

2. An investor should purchase a stock when

A) the market price exceeds the intrinsic value.
B) the expected rate of return equals or exceeds the required return.
C) the capital gains rate is less than the required return and no dividends are paid.
D) the market price is greater than the justified price.

3. Heather believes that by carefully examining a company's fundamentals and by applying the best valuation models she can identify stocks whose market prices are lower than their intrinsic values. In order for this to be true

A) she needs an accurate estimate of future earnings and dividends.
B) some stocks must be incorrectly priced.
C) betas must be stable over time.
D) P/E ratios for both the stock and the market must be stable over time.

4. For a stock investment, the dividend yield is calculated by

A) dividing a stock's annual cash dividend by its price.
B) dividing a stock's price by its annual cash dividend.
C) multiplying a stock's semi-annual dividend by two.
D) dividing the annual change in the stock's price plus its annual dividend amount by the beginning of the year price.

5. NZMA stock is currently selling for $128. Which of the following options is "in-the-money"?

A) March 130 call
B) February 125 call
C) March 125 put
D) February 100 put

Question 2: Answer the following questions: (25 marks)

Contrast between dollar-cost and constant-dollar plans in the context of portfolio management (25 marks, 12.5 marks for each plan).

Question 3: Answer the following questions: (25 marks)

A. Last year, Black Hole Company paid $90 million of interest expense, and its average rate of interest for the year was 15%. The company's ROE is 20%, and it pays no dividends. Estimate next year's interest expense, assuming that interest rates will fall by 20% and the company keeps a constant equity multiplier of 25%. (15 marks).
B. In 2018, Peugeot had a ROE of 14.4%. Its earnings per share were 3.16, and its dividends per share were €0.97. Estimate Peugeot's growth rate. (10 marks).

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Question 4: Answer the following questions: (25 marks)

A. Lew paid $600 to purchase a call on Delta stock with a strike price of $10. What does the market price of Delta have to be for Lew to break-even on his option investment? Ignore transaction costs and taxes (10 marks).

B. The risk-free rate is currently 8.1%. Use the data in the accompanying table for the Fio family's portfolio and the market portfolio during the year just ended to answer the questions that follow (15 marks):

Data Item	Fios' Portfolio	Market Portfolio
Rate of return	12.8%	11.2%
Standard deviation of return	13.5%	9.6%
Beta	1.10	1.00

1. Calculate Sharpe's measure for the portfolio and the market. Compare the two measures, and assess the performance of the Fios' portfolio during the year just ended. (5 marks).
2. Calculate Treynor's measure for the portfolio and the market. Compare the two measures, and assess the performance of the Fios' portfolio during the year just ended. (5 marks).
3. Calculate Jensen's measure (Jensen's alpha). Use it to assess the performance of the Fios' portfolio during the year just ended. (5 marks).

"End of Exam"

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